Linked List

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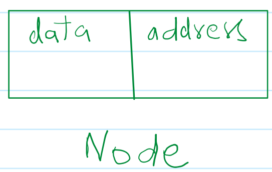
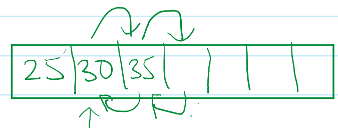
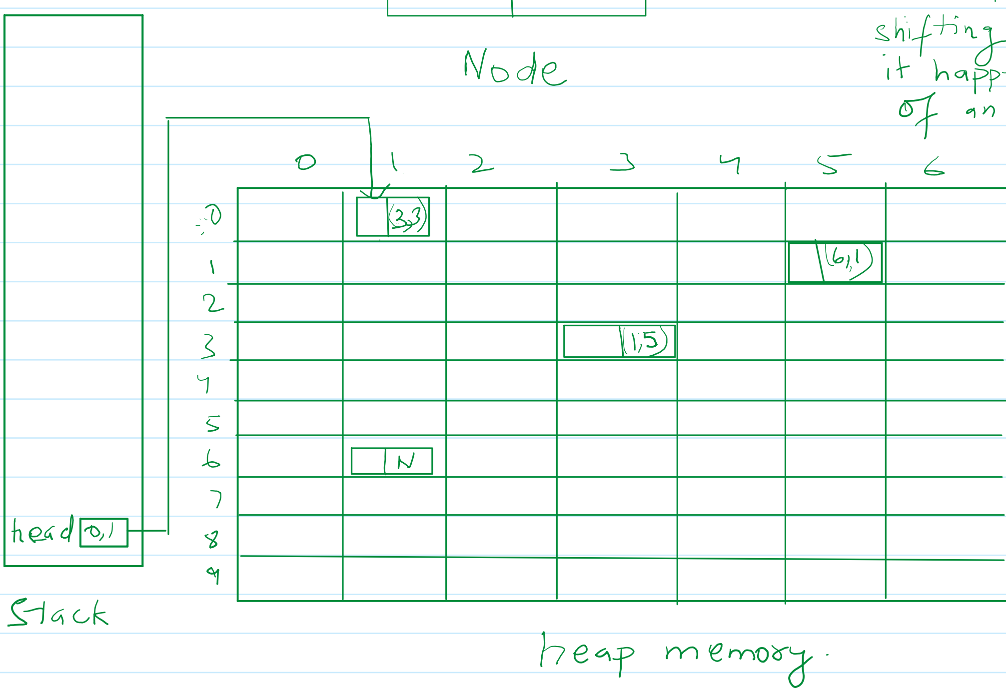
A linked list is a linear data structure that includes series of connected nodes. A linguist can be defined as a set of nodes that are randomly stored in memory. A node in the linked list contains 2 parts. That is the first is the data part and the 2nd is the address part. The last node of the list contains a pointer to NULL. After array Linked list is the second most used data structure. In a linked list, every link contains a connection to another link.



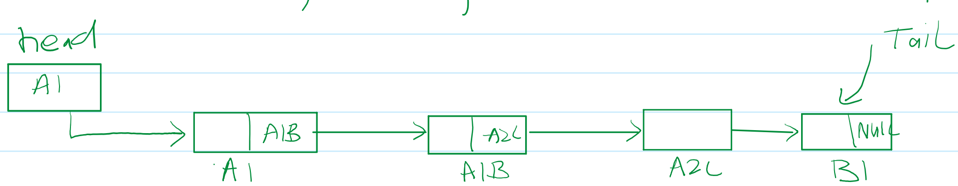


shifting elements in array is time consuming. it happens during insertion or deletion of an element.



What is a Node?

A node is a self referential structure / class. It means, it can hold the reference of another node in it of its own type.



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How to declare a Node?

```
struct Node {

    int data;
    struct Node *next;
};
```

```
    int data;
    struct Node * next;
};
```

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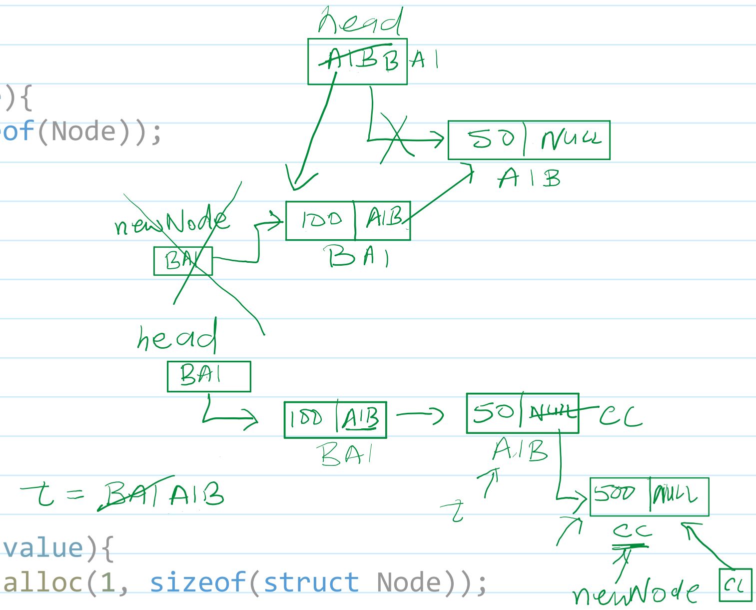
```
void addNodeAtStart(Node **head, int value){
    Node *newNode = (Node *)calloc(1, sizeof(Node));
    newNode->data = value;
    newNode->next = NULL;
    if(*head == NULL){
        *head = newNode;
        return;
    }
    newNode->next = *head;
    *head = newNode;
}
```




```
void addNodeAtEnd(struct Node **head, int value){
    struct Node*newNode = (struct Node *)calloc(1, sizeof(struct Node));
    newNode->data = value;
    newNode->next = NULL;
    if(*head == NULL){
        *head = newNode;
        return;
    }
    struct Node *t = *head;
    for(;t->next;t = t->next);
    t->next = newNode;
}
```